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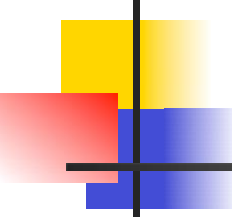
Lecture 7

Chapter 1. The Foundations

1.6 Introduction to Proofs

Proof Terminology

- A ***proof*** is a valid argument that establishes the truth of a mathematical statement
- ***Axiom*** (or ***postulate***): a statement that is assumed to be true
- ***Theorem***
 - A statement that has been proven to be true
- ***Hypothesis, premise***
 - An assumption (often unproven) defining the structures about which we are reasoning

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More Proof Terminology

- ***Lemma***

- A minor theorem used as a stepping-stone to proving a major theorem.

- ***Corollary***

- A minor theorem proved as an easy consequence of a major theorem.

- ***Conjecture***

- A statement whose truth value has not been proven. (A conjecture may be widely believed to be true, regardless.)

Proof Methods

- For proving a statement p alone
 - ***Proof by Contradiction*** (indirect proof):
Assume $\neg p$, and prove $\neg p \rightarrow \mathbf{F}$.

Proof Methods

- For proving implications $p \rightarrow q$, we have:
 - **Trivial proof:** Prove q by itself.
 - **Direct proof:** Assume p is true, and prove q .
 - **Indirect proof:**
 - **Proof by Contraposition** ($\neg q \rightarrow \neg p$):
Assume $\neg q$, and prove $\neg p$.
 - **Proof by Contradiction:**
Assume $p \wedge \neg q$, and show this leads to a contradiction. (i.e. prove $(p \wedge \neg q) \rightarrow \mathbf{F}$)
 - **Vacuous proof:** Prove $\neg p$ by itself.

Direct Proof Example

- **Definition:** An integer n is called *odd* iff $n=2k+1$ for some integer k ; n is *even* iff $n=2k$ for some k .
- **Theorem:** Every integer is either odd or even, but not both.
 - This can be proven from even simpler axioms.

- **Theorem:**
(For all integers n) If n is odd, then n^2 is odd.

Proof:

If n is odd, then $n = 2k + 1$ for some integer k .

Thus, $n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$.

Therefore n^2 is of the form $2j + 1$ (with j the integer $2k^2 + 2k$), thus n^2 is odd. ■

Indirect Proof Example: Proof by Contraposition

- **Theorem:** (For all integers n)
If $3n + 2$ is odd, then n is odd.

- **Proof:**

(Contrapositive: If n is even, then $3n + 2$ is even)

Suppose that the conclusion is false, *i.e.*, that n is even.

Then $n = 2k$ for some integer k .

Then $3n + 2 = 3(2k) + 2 = 6k + 2 = 2(3k + 1)$.

Thus $3n + 2$ is even, because it equals $2j$ for an integer $j = 3k + 1$. So $3n + 2$ is not odd.

We have shown that $\neg(n \text{ is odd}) \rightarrow \neg(3n + 2 \text{ is odd})$,
thus its contrapositive $(3n + 2 \text{ is odd}) \rightarrow (n \text{ is odd})$ is
also true. ■

Vacuous Proof Example

- Show $\neg p$ (i.e. p is false) to prove $p \rightarrow q$ is true.
- **Theorem:** (For all n) If n is both odd and even, then $n^2 = n + n$.
- **Proof:**

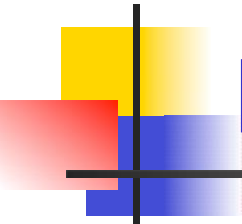
The statement “ n is both odd and even” is necessarily false, since no number can be both odd and even. So, the theorem is vacuously true. ■

Trivial Proof Example

- Show q (i.e. q is true) to prove $p \rightarrow q$ is true.
- **Theorem:** (For integers n) If n is the sum of two prime numbers, then either n is odd or n is even.
- **Proof:**
Any integer n is either odd or even. So the conclusion of the implication is true regardless of the truth of the hypothesis. Thus the implication is true trivially. ■

Proof by Contradiction

- A method for proving p .
 - Assume $\neg p$, and prove both q and $\neg q$ for some proposition q . (Can be anything!)
 - Thus $\neg p \rightarrow (q \wedge \neg q)$
 - $(q \wedge \neg q)$ is a trivial contradiction, equal to **F**
 - Thus $\neg p \rightarrow \mathbf{F}$, which is only true if $\neg p = \mathbf{F}$
 - Thus p is true

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Rational Number

- Definition:

The real number r is *rational* if there exist integers p and q with $q \neq 0$ such that $r = p/q$.
A real number that is not rational is called *irrational*.

Proof by Contradiction Example



■ **Theorem:** $\sqrt{2}$ is irrational.

■ **Proof:**

■ Assume that $\sqrt{2}$ is rational. This means there are integers x and y ($y \neq 0$) with no common divisors such that $\sqrt{2} = x/y$.

Squaring both sides, $2 = x^2/y^2$, so $2y^2 = x^2$. So x^2 is even; thus x is even (see earlier).

Let $x = 2k$. So $2y^2 = (2k)^2 = 4k^2$. Dividing both sides by 2, $y^2 = 2k^2$. Thus y^2 is even, so y is even.

But then x and y have a common divisor, namely 2, so we have a contradiction.

Therefore, $\sqrt{2}$ is irrational. ■

Proof by Contradiction

- Proving implication $p \rightarrow q$ by contradiction
 - Assume $\neg q$, and use the premise p to arrive at a contradiction, i.e. $(\neg q \wedge p) \rightarrow \mathbf{F}$
 $(p \rightarrow q \equiv (\neg q \wedge p) \rightarrow \mathbf{F})$
 - How does this relate to the proof by contraposition?
 - ***Proof by Contraposition*** $(\neg q \rightarrow \neg p)$:
Assume $\neg q$, and prove $\neg p$.

Proof by Contradiction

Example: Implication



- **Theorem:** (For all integers n)
If $3n + 2$ is odd, then n is odd.

- **Proof:**

Assume that the conclusion is false, *i.e.*, that n is even, and that $3n + 2$ is odd.

Then $n = 2k$ for some integer k and $3n + 2 = 3(2k) + 2 = 6k + 2 = 2(3k + 1)$. Thus $3n + 2$ is even, because it equals $2j$ for an integer $j = 3k + 1$.

This contradicts the assumption “ $3n + 2$ is odd”.

This completes the proof by contradiction, proving that if $3n + 2$ is odd, then n is odd. ■

Circular Reasoning

- The fallacy of (explicitly or implicitly) assuming the very statement you are trying to prove in the course of its proof. Example:
- Prove that an integer n is even, if n^2 is even.
- **Attempted proof:**

Assume n^2 is even. Then $n^2 = 2k$ for some integer k .

Dividing both sides by n gives $n = (2k)/n = 2(k/n)$.

So there is an integer j (namely k/n) such that $n = 2j$.
Therefore n is even.

- Circular reasoning is used in this proof.
Where?

*Begs the question: How do you show that $j = k/n = n/2$ is an integer, without **first** assuming that n is even?*